

The empirical recurrence for OEIS A231045, proved by transfer matrix

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Abstract

OEIS A231045 counts arrays of width 4 over $\{0, 1, 2, 3\}$ in which every cell must see, among its neighbours in named directions, a stated number carrying the value one MORE than its own modulo 4, and in some of the family a stated number carrying the value one less. The entry carries an empirical recurrence of order 3 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: a cell's neighbours lie in three consecutive rows, so the count is a walk count on $S = 1865$ states.

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1 The conjecture

OEIS A231045 is “Number of $n \times 4$ $0..3$ arrays $x(i,j)$ with each element horizontally, diagonally or antidiagonally next to at least one element with value $(x(i,j)+1) \bmod 4$ and at least one element with value $(x(i,j)-1) \bmod 4$, no adjacent elements equal, and upper left element zero.”. It has offset 1 and begins

$$0, 32, 128, 2080, 18688, 209440, 2150784, 22847008, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

$$\text{Empirical: } a(n) = 8*a(n-1) + 33*a(n-2) - 68*a(n-3).$$

As of the “Last modified” line on the live entry (Sep 25 2018 14:50:41, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array has 4 cells across, with entries in $\{0, 1, 2, 3\}$. The NEIGHBOURS of a cell are the cells reached from it by the offsets

$$(-1, -1), (-1, 1), (0, -1), (0, 1), (1, -1), (1, 1),$$

those lying inside the array. Writing v for the value of a cell, the condition is that among its neighbours there be at least 1 carrying $v + 1 \bmod 4$ and at least 1 carrying $v - 1 \bmod 4$. The entry also forbids two neighbouring cells from carrying equal values. The top left cell is fixed at 0.

Note that the values and the modulus agree: the alphabet has 4 letters and the arithmetic is modulo 4, so “one more than v ” wraps from 3 round to 0.

3 Three rows decide one

Lemma 1. *Every offset above changes the row index by -1 , 0 or $+1$, so all the neighbours of a cell of row i lie in rows $i-1$, i and $i+1$, and the condition for the whole of row i is decided by those three rows. A neighbour outside the array is not a neighbour and contributes nothing to any of the counts.*

Proof. Immediate from the offsets listed. $\square$

So the state is the pair (row above, current row), the row above allowed to be the symbol $\perp$, and a step appends a row and settles the condition for the middle row of the window it completes. Two boundaries are decided rather than assumed. The rows that may BEGIN an array are exactly those whose first entry is 0 , which is the top-left clause. And the condition for the LAST row is settled by the end vector, with nothing below it: the end vector is therefore not the all-ones vector, and a row that sees enough neighbours only when a further row is written below it does not finish an array. With M the adjacency matrix,

$$a(n) = \iota^\top M^j \tau, \quad S = 1865.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C -finite.

4 The proof

Lemma 2. *Let $q(t) = t^3 - \sum_i c_i t^{3-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+3} - \sum_i c_i M^{j+3-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 3.

$$a(n) = 8a(n-1) + 33a(n-2) - 68a(n-3)$$

for every $n > 3$.

Proof. The vector $w = q(M)\tau$ was formed by 3 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 3 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. The leading zeros the entry publishes are a sharp test of the boundary: a small array has too few neighbours for any cell to see what the condition demands.

Second, the count was recomputed for the smallest arrays straight from the definition: enumerate every array of that size in the orientation the entry's own name writes, count each cell's neighbours of each kind, and keep the array when every cell is satisfied. No window, no transfer matrix and no transposition are involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.